

Jonas Ngnawé

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Google Scholar

Education

Université Laval, Mila - Quebec AI Institute, Stanford University (VSR) <i>PhD in Computer Science: Adversarial Machine Learning</i>	Quebec City, Canada 2021–2027
AIMS African Master's of Machine Intelligence <i>MSc in Machine Learning</i> Funded by Meta and Google; taught by world top researchers.	Kigali, Rwanda 2018–2019
African Institute for Mathematical Sciences (AIMS) <i>MSc in Mathematical Sciences</i>	Limbe, Cameroon 2016–2017
École Polytechnique <i>MSc in Computer Engineering</i>	Yaoundé, Cameroon 2011–2016
Summer Schools: Machine Learning Summer School (MLSS), London 2019 ; Deep Learning & Reinforcement Learning Summer School (DLRL), Edmonton 2025 .	

Publications

- RobustGenBench: A Benchmark for Robust Generalization to Adversarial and Common Perturbations, with Applications to Vision and Vision-Enabled Large Language Models, *Heuillet, M., Ngnawé, J., Bhagwatkar, R., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A. (Under Review)*
- Robust Fine-Tuning from Non-Robust Pretrained Models: Mitigating Suboptimal Transfer with Epsilon-Scheduling, *Ngnawé, J., Heuillet, M., Sahoo, S., Pequignot, Y., Ahmad, O., Durand, A., Précioso, F., & Gagné, C. (NeurIPS 2025 Workshop on Reliable ML, ICLR 2026). [OpenReview]*
- Detecting Brittle Decisions for Free: Leveraging Margin Consistency in Deep Robust Classifiers, *Ngnawé, J., Sahoo, S., Pequignot, Y., Précioso, F., & Gagné, C. (NeurIPS 2024). [NeurIPS Proceedings]*
- A Layer Selection Approach to Test Time Adaptation, *Sahoo, S., ElAraby, M., Ngnawé, J., Pequignot, Y., Précioso, F., & Gagné, C. (AAAI 2025). [AAAI Proceedings]*
- A Guide to Robust Generalization: The Impact of Architecture, Pre-training, and Optimization Strategy, *Heuillet, M., Bhagwatkar, R., Ngnawé, J., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A. (NeurIPS 2025 Workshop on Reliable ML). [arXiv]*
- Robustmix: Improving Robustness by Regularizing the Frequency Bias of Deep Networks, *Ngnawé, J., Njifon, M. A., Heek, J., & Dauphin, Y. (NeurIPS 2022 Workshop on Distribution Shifts). [OpenReview]*
- Scalable Inference with Distributed Gaussian Processes using the Wasserstein Barycenter, *Ngnawé, J., Deisenroth, M. (Master's project). [GitHub]*
- k-Clustering Problem: Application to a Field Technician Allocation Optimization Problem, *Ngnawé, J., Zacko, E., Wandji, N., & Ndjinga, M. (Master's project). [ResearchGate]*

AI/ML Work Experience

Vector Institute <i>Machine Learning Associate</i>	Toronto, Canada May – Aug 2026
○ FastLane MLA Program: Delivering real-world AI implementation projects with Canadian startups. Build deployment-ready AI proof-of-concepts with emphasis on documentation and stakeholder presentation.	
○ ML Inference optimization at AutoInference.ai.	

Stanford Artificial Intelligence Lab (SAIL)

Visiting Student Researcher

Palo Alto, USA

Oct 2025 – June 2026

- Safety, Robustness and Uncertainty Estimation in (Multimodal) Vision-Language Models.
- AI Measurement Science applied to LLMs Evaluation

Google Brain

Google AI Resident

Accra, Ghana

Oct 2019 – Sep 2021

- Development of robust methods for deep neural networks.
- Compression of deep neural networks by quantization and pruning.

Other Work Experience

GroupOne Holding Company

Computer Engineer

Limbe, Cameroon

Jan – Aug 2018

- Website development; fuel consumption prediction projects.
- Studies for the creation of a startup incubator (business plan).
- Industrial maintenance (refinery) of the Honeywell Experion System; IT support.

Tools: HTML5/CSS3/JavaScript/jQuery/jQueryUI/Ajax/PHP, MySQL, PMI Methodology, Business Model Canvas.

Freelance

Software Developer

Cameroon

2017–2018

- Digitization of administrative procedures for various ministries in Cameroon.
- Development and maintenance of WordPress plugins (WooCommerce).

Tools: DevOps, Electronic Document Management (EDM)/Electronic Archiving (Maarch RM), PHP, WooCommerce, MySQL Workbench, Microsoft Access, Git/GitLab/GitHub, Java, NetBeans, SCRUM.

Bolloré Africa Logistics

Software Developer Intern

Douala, Cameroon

Feb – Aug 2016

- Automation and analysis of the review of access rights to the Information System.
- Detection of abnormal access rights.

Tools: Talend Open Studio, QlikView, C#, ASP.NET, Microsoft SQL Server, Visual Studio, UML/Merise, BPMN 2.0, PMI Methodology.

Talks

Host at Kmer AI Talks.

2024: Panelist, “AI Regulation & Fairness in the Generative AI Era,” Black in AI Workshop @ NeurIPS.

2024–2025: “Detecting Brittle Decisions for Free” — Kmer AI (Jun 2024); Galsen AI (Jan 2025).

2022: Keynote, “Robustness in AI,” Deep Learning IndabaX-Cameroon (Dec 27, 2022).

Reviews

NeurIPS, ICLR, AAAI; IEEE Transactions on Image Processing.

Algorithmic Fairness workshops @ NeurIPS 2021, 2022, 2023, 2024.

Montreal AI Symposium (2022, 2024); Black in AI Workshop (2022–2024).

Deep Learning Indaba, Pan-African Machine Learning Conference (2023).

Leadership & Event Organization

Organizing committee: Interpolate Workshop @ NeurIPS 2022 (First Workshop on Interpolation Regularizers and Beyond).

Co-organizer: Machine Learning Projects Competition (MLPC, 2000 USD) 2020–2022, Polytechnique Yaoundé.

Co-organizer: Two-week introductory course in Machine Learning (Sep 2019), Polytechnique Yaoundé.

Co-organizer: First Deep Learning IndabaX Rwanda (Apr 2019), Kigali, Rwanda.

Co-Founder of Kmer-AI, an association for promoting and popularizing AI and ML in Cameroon (since 2020).

AI/ML Skills

ML Frameworks: PyTorch, TensorFlow, Scikit-Learn

MLOps Tools: Git/GitHub, Weights & Biases, Docker, Slurm, Google Cloud Platform, vLLM, Alperf

Other: Training Large VLMs from scratch , Robust Evaluation LLMs/LVLMs